Assignment 4

"Take one domain and build business Understanding"

Domain: Banking Industry (Credit Card Fraud Detection)

BUSINESS PROBLEM:

Banks issue credit cards to customers. Some customers or external attackers perform fraudulent transactions using stolen or misused cards, causing significant financial losses and reputational damage to the bank.

1. Business Objective

- Identify potential fraudulent credit card transactions in real time
- Reduce financial losses due to fraud
- Improve transaction monitoring and risk management
- Flagging suspicious transactions for manual review
- Strengthening transaction authentication

Business Success Criteria

- Reduction in fraudulent transactions (e.g., by 30%)
- Faster fraud detection time
- Lower financial losses
- Improved customer trust and security

2. Assess Situation

Inventory of Resources

- Historical transaction data
- Customer account information
- Device and location data
- Fraud investigation team
- Existing banking IT infrastructure

Requirements

- Highly accurate fraud detection
- Real-time or near real-time prediction
- Model explainability for regulatory compliance
- Minimal disruption to genuine customers

Assumptions

- Historical fraud patterns can predict future fraud
- Transaction data captured by the bank is reliable
- Fraudulent behavior shows detectable patterns

Constraints

- Strict financial regulations and data privacy laws
- Highly imbalanced dataset (fraud is rare)
- Need for very low latency predictions
- Limited labeled fraud examples

Costs and Benefits

Costs:

- Infrastructure and data storage
- Model development and maintenance
- Skilled data science team
- Real-time monitoring systems

Benefits:

- Reduced fraud losses
- Improved customer confidence
- Better risk management
- Increased bank profitability

3. Determine Data Science Goals

Data Science Objective (Technical View)

Build a classification model to predict whether a transaction is:

- Fraudulent (Yes)
- Genuine (No)

Data Science Tasks

- Data cleaning and preprocessing
- Feature engineering (transaction amount, location change, transaction frequency, time patterns)
- Handling missing values and class imbalance
- Train models such as:
    - Logistic Regression
    - Decision Tree
    - Random Forest
    - XGBoost

Data Science Success Criteria

- High overall accuracy
- High Recall for fraud cases (critical)
- Good Precision (avoid blocking genuine users)

- AUC-ROC score above 0.90

Produce Project Plan

Stage No	Stage Name		Main Activities		Duration
1	Business Understanding		Define problem, objectives, success criteria		1 Week
2	Data Collection & Understanding		Gather and explore transaction data		2 Weeks
3	Data Cleaning & Preparation		Handle missing values, feature engineering		2 Weeks
4	Model Building		Train ML models		3 Weeks
5	Model Evaluation		Evaluate using Accuracy, Recall, Precision, AUC		1 Week
6	Deployment		Deploy model into banking system		1 Week
7	Monitoring & Maintenance		Monitor performance and retrain		Continuous

Resources Needed

- Data Scientists
- Banking domain experts
- IT support team
- Database administrators

Tools & Techniques

- Python (Pandas, NumPy, Scikit-learn)
- SQL for data extraction
- Power BI / Tableau for dashboards
- Machine learning classification algorithms

Final Outcome

- A Credit Card Fraud Detection System
- Detects fraud in near real time
- Reduces financial losses
- Improves customer security
- Aligns data science goals with business objectives